

Virtual Reality Learning Optimization: Immersive Debriefing with Review and Redo for Effective, Targeted Training

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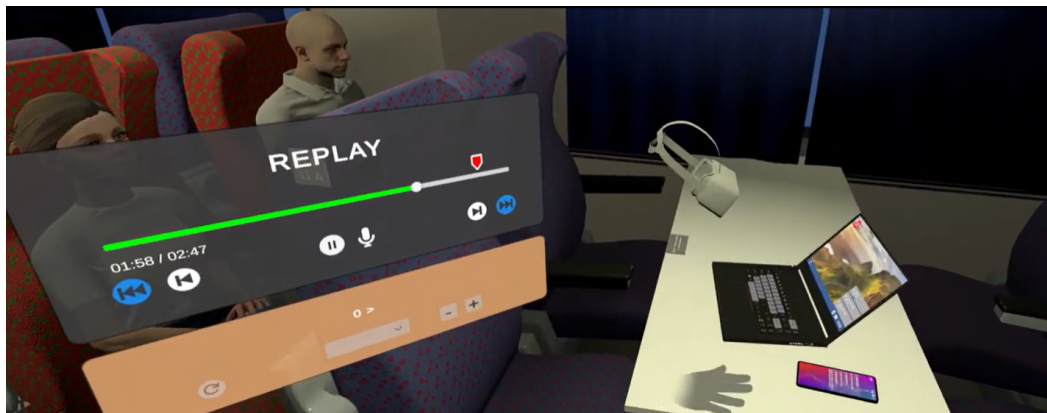


Figure 1: Interface Review/Redo system

ABSTRACT

In VR training, debriefing is as crucial as the simulation phase. With the growing adoption of these pedagogical tools, defining optimal educational approaches to maximize benefits for trainers and learners becomes essential. However, despite their benefits, VR-adapted debriefing methods still need to be explored. This paper presents an adaptable, all-in-one immersive debriefing module. It includes a complete system for recording, reviewing, and redoing actions. An ongoing study explores the redo module's effect in dynamic training scenarios. This module could enhance learning and reinforce immersive debriefing systems' interest and relevance.

Index Terms: Debriefing, Simulation, Review, Redo, Learning

1 INTRODUCTION

The growing use of VR in training, particularly in healthcare, enables users to acquire unique experiences while offering real-time monitoring. This increases learner motivation and commitment. VR simulators are used in sectors such as medicine and the military to expose learners to risky situations without any real danger.

Debriefing is crucial in VR training, allowing learners to memorize and interpret their experiences and acquire new skills [5]. However, despite their pedagogical benefits, few studies have focused on specific VR debriefing methods. The main question is: **can we propose a design method for an immersive debriefing environment suitable for VR simulators?** Our aim is to develop a debriefing

environment adapted to VR simulators, integrating various pedagogical approaches and simulation specificities.

2 STATE OF THE ART SUMMARY

The term 'debriefing' originated in the military context to gather post-mission information. It has been extended to the psychological evaluation of events and is now applied to various fields such as education and psychology. According to the World Health Organization, debriefing is an immediate feedback session after an exercise where participants share their perspectives on strengths, weaknesses, and areas of improvement and where these contributions are integrated into the exercise report.

A successful debriefing includes the participants, the simulation experience, the impact of the simulation, the debriefing materials, the timing, and the role of the facilitator. The ideal time for debriefing is at the end of the simulation to maximize emotional and clinical realism. The facilitator guides the discussion, helping participants to process and reflect on their experiences [6].

Various debriefing techniques exist, with the RAS (Reaction, Analyze, Summarize) [1] method being the most commonly used. It consists of three phases: Reaction, where participants express their post-training emotions; Analysis, examining the key events and participants' states of mind; and Summary, assessing the skills acquired. This structure, common to most debriefing techniques, is essential for designing our immersive system.

Few studies have examined immersive debriefing, particularly in medicine. Current debriefing methods are generally oral or written, with the occasional use of video for analysis. However, 360° videos currently represent the most immersive method for revisiting actions performed during the learning session. Nicholson et al. [4] explored the potential benefits of VR in the debriefing process. Participants noted that VR provided a more immersive debriefing environment, leading to greater engagement in their performance and a better understanding of simulation events.

Several studies have explored the implementation of immersive replay systems to enhance the learning experience by making it more engaging and interactive [2]. These systems enable learners to

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revisit and dynamically analyze their actions in detail. In one study, participants reported feeling more engaged and immersed when using an immersive multi-user replay system compared to traditional methods, such as co-viewing 360° videos. This highlights the potential of immersive systems to enrich the debriefing process and facilitate deeper reflection and better information retention.

Existing VR debriefing solutions often place learners in a predominantly passive role, which can hinder active engagement and critical reflection. Moreover, prolonged sessions may exacerbate eye fatigue, diminishing engagement and focus. Another significant drawback is the difficulty in perceiving non-verbal cues—essential for effective debriefing—when participants are confined to headsets. These limitations underline the need for a more interactive and learner-centered design of debriefing systems. Our objectives are to improve learners' motivation and information retention. By creating a more engaging environment and facilitating active participation, we aim to maximize the debriefing effectiveness and promote better assimilation of information.

3 SUGGESTED SOLUTION

We present a system designed to enhance debriefing sessions following VR simulations. Its primary goal is to provide immersive, detailed feedback on participants' performance, allowing them to revisit key simulation moments directly within the virtual environment (See Fig. 1).

The system includes a replay function that enables learners to review their actions in a third-person perspective using an avatar. This perspective enhances immersion while reducing the disorientation often associated with first-person views. Building on this, the system incorporates an innovative redo feature that allows learners to restart the simulation from any selected point during the review phase to explore alternative actions. When a redo is activated, simulation objects are restored to their previous state using precise recordings or interpolations, ensuring seamless continuity. New actions are recorded in a separate segment, creating distinct timelines that can be compared during subsequent review phases (see Fig. 2). This structured experimentation enhances learning by allowing learners to refine their strategies, explore outcomes, and achieve mastery through targeted practice.

The system is compatible with Unity 3D-based simulators and uses an optimized recording architecture to efficiently capture and restore only the modified states of objects and components. Common Unity features, such as object transformations, lighting, and audio sources, are automatically detected and recorded, simplifying integration for developers. This adaptability ensures that the system can be easily incorporated into a wide range of simulation environments.

4 EVALUATION OF THE SYSTEM

Our initial study, involving 36 participants, compared an immersive debriefing system with traditional discussion-based and video-assisted debriefing approaches [3]. Results demonstrated that immersive debriefing was as usable and memorable as traditional methods in a linear scenario. Building on this, our ongoing study evaluates the added value of the redo feature [7]. This study involves two groups of learners in nomadic work training guided by the same cybersecurity trainer. The first group uses a replay-only system, while the second group employs an immersive debriefing system with both replay and redo functionalities. A cybersecurity simulator in a railway context serves as the study environment.

The evaluation focuses on the impact of the redo feature on motivation, usability, and information retention. By enabling targeted repetition, redo allows learners to experiment with alternative strategies, analyze their consequences, and refine their decision-making without replaying the entire simulation. This approach

aims to improve debriefing quality, enhance learner engagement, and strengthen retention of acquired skills.

The system's ability to let learners relive their actions in an immersive environment fosters critical reflection and performance analysis. By supporting revision, experimentation, and focused correction, the system would optimize learning efficiency and would facilitate mastery. This makes it particularly valuable in healthcare and crisis management contexts, where learners can safely practice procedures, test strategies, and adapt responses through iterative practice. Integration with Virtual Humans and real-time trainer feedback further enhances the learning experience by providing personalized guidance and fostering deeper retention of skills.

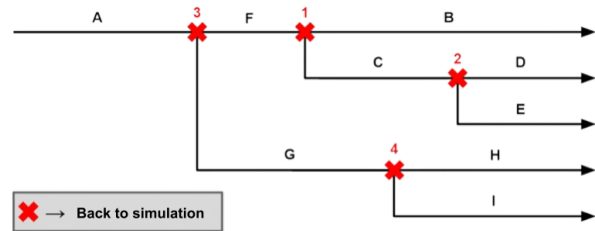


Figure 2: Illustration of Redo Functionality with Timeline Splits in Simulation

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